Explainable AI for the Sift Cross-Encoder

Token-level and line-level attribution with Integrated Gradients, a deletion-based faithfulness check and worked cases

PROJECT	DELIVERABLE	TEAM NO	TEAM MEMBER	METHOD
Sift	**Deliverable 1: XAI**	**Team 24 (single member)**	**Kundhave S, CB.SC.U4CSE23037**	**Integrated Gradients (Captum)**

1. Executive Summary

Sift reads a developer intent and one diff hunk jointly and outputs P(misaligned). The XAI layer answers one question: which intent words and which diff lines pushed that probability up or down. It uses Integrated Gradients (IG) on the ModernBERT cross-encoder, aggregated from sub-tokens to words (intent and hunk) and to diff lines (hunk), and shows the result in the review UI.

- **Method:** IG with a [PAD]-embedding baseline, 24 integration steps in the experiments (the UI default is 16), target = sigmoid(alignment logit) = P(misaligned); Input x Gradient is run as a cheaper comparator.
- **Faithfulness (deletion test, n = 100 test hunks):** replacing the top-5 IG tokens with [PAD] lowers mean P(misaligned) from 0.579 to 0.386, against 0.482 when 5 random tokens are masked; the top-5 set beats random in 66% of hunks. The random-masking drop is large too, so part of the effect is an artefact of masking itself.
- **Cost:** one IG explanation takes p50 506 ms (p95 1328 ms) on the GPU, against p50 24.7 ms for scoring alone; it is therefore computed on demand for the top-ranked hunks, not for every hunk.
- **Cases:** one real test-split example per class (Aligned, Collateral, Scope-Expansion, Purpose-Drift) are shown with highlighted tokens. They were chosen as the most confident correct short examples, so they are illustrative, not representative.
- **Limit:** attribution is evidence about the model, not proof of the cause of a misalignment.

2. Mathematical Formulation

Let x be the embedded input sequence (intent and hunk, joined as in training), x' the baseline sequence and F(x) = $\sigma(z_{align}(x))$ the probability of the misaligned class. For embedding dimension j of token t, Integrated Gradients is the path integral of the gradient from the baseline to the input:

$$\mathrm{IG}_{t,j}(x) = (x_{t,j} - x'_{t,j}) \int_0^1 \frac{\partial F(x' + \alpha\,(x - x'))}{\partial x_{t,j}}\, d\alpha$$

The integral is approximated by a Riemann sum with m = 24 steps (m = 16 in the UI):

$$\mathrm{IG}_{t,j} \approx (x_{t,j} - x'_{t,j})\, \frac{1}{m} \sum_{k=1}^{m} \frac{\partial F(x' + \frac{k}{m}(x - x'))}{\partial x_{t,j}}$$

The token score sums over embedding dimensions, and the completeness property (scores add up to F(x) - F(x')) is what makes IG attractive; Input x Gradient is the one-step version:

$$a_t = \sum_j \mathrm{IG}_{t,j} \qquad \mathrm{IxG}_{t,j} = x_{t,j}\, \frac{\partial F(x)}{\partial x_{t,j}}$$

Baseline. Every content token (intent and hunk) is replaced by the embedding of the [PAD] token; special tokens ([CLS], [SEP]) keep their own embedding so the sequence structure is unchanged. **Aggregation.** Sub-token scores are summed to words by character offsets (a word is an identifier or one punctuation mark); for hunks they are also summed per diff line. The sign is kept: positive (red) pushes toward misaligned, negative (blue) toward aligned. The mean share of absolute attribution that falls on intent tokens is 19.0% (hunks carry the rest, which matches their larger share of the input).

3. Implementation Notes

- **Code:** src/sift/explain.py (class built on Captum IntegratedGradients and InputXGradient, attributions taken at the embedding layer; word and line aggregation; deletion test), experiments/xai.py (experiment driver), the FastAPI endpoint /api/analyze with *explain: true* and the Review page.
- **Numerics:** attributions were computed with bf16 weights, internal batch 2 and a 1.5 GB GPU memory cap (fp32 ran out of memory under the cap), so the numbers describe the bf16 model.
- **Faithfulness protocol:** 100 test hunks predicted misaligned at the validation-tuned threshold 0.34 (49% of them truly misaligned); for k in 3, 5, 10, 20 the k highest-scoring content tokens are replaced by [PAD] and compared with k random content tokens (RandomState 473).
- **Scope:** Integrated Gradients and Input x Gradient are implemented; attention rollout (optional) is outside this work.

4. Attribution Catalogue

Item	Level	Role	Status
Integrated Gradients	sub-token, word, diff line	Primary explanation shown in the UI	implemented and measured
Input x Gradient	sub-token	Cheap comparator in the faithfulness test	implemented and measured
Attention rollout	sub-token	Optional	not implemented
Word aggregation (intent)	word	Which request words drove the verdict	implemented
Line aggregation (hunk)	diff line	Which changed lines drove the verdict	implemented
Intent attribution share	scalar per hunk	Mean 19.0% of |attribution| on intent tokens	measured
Deletion faithfulness	top-k vs random-k masking	Sanity check of the attribution ranking	measured, n = 100

5. UI Presentation

The Review page ranks hunks by calibrated P(misaligned). Ticking *Explain top hunks* adds, for the selected hunk, the intent words and the diff lines coloured by IG score (the right panel below), with the caveat that it is evidence about the model, not proof of cause. Highlights are only computed on request because of the latency cost in section 8.

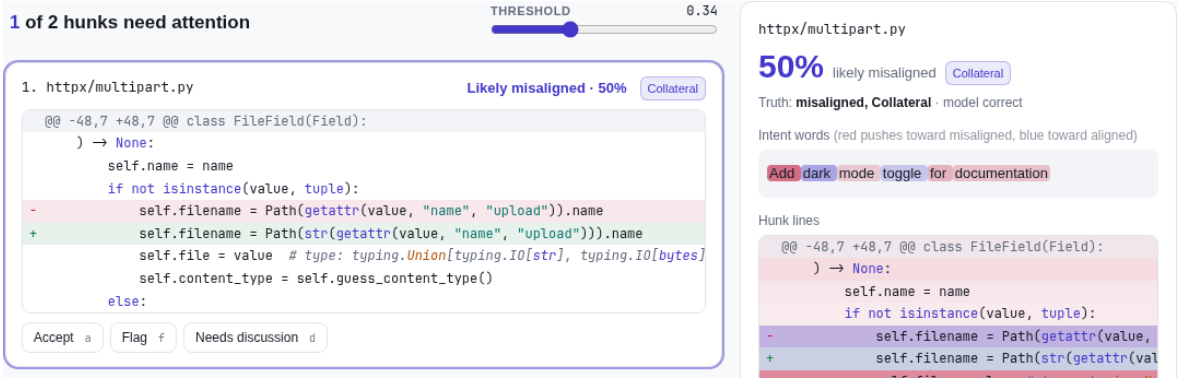


Figure 1. Screenshot of the running app (Review page, test-set example, light theme): hunk card and the Explain panel with word and line highlights.

6. Worked Case Studies (test split)

One example per class from *results/xai_cases_ui.json*. Selection rule (experiments/xai.py): the most confident correct prediction with a short hunk, so these are favourable examples. Red marks pushed P(misaligned) up, blue pushed it down; colour intensity is relative to the largest absolute score within that segment.

Class	Sample	P(misal.)	Top intent word	Top hunk line (IG score)
Aligned	SIFT-017909 (attrs)	0.025	latest (-0.065)	+ - uses: "actions/setup-python@v2" (+0.112)
Collateral	SIFT-017393 (attrs)	0.957	. (+0.153)	hints = typing.get_type_hints(cls, globalns=globalns, localn (+0.169)
Scope-Expansion	SIFT-017837 (axios)	0.898	isStandardBrowserEnv (+0.285)	/** (+0.739)
Purpose-Drift	SIFT-017387 (httpx)	0.908	the (+0.248)	+ "type": "http.response.start", (+0.219)

Aligned | true: aligned | P(misaligned) = 0.025 | attrs, SIFT-017909

Intent (word-level IG)

Use latest setup - python

Hunk lines (line-level IG: sum of token attributions per line)

```
.github/workflows/main.yml                                                +0.03
@@ -23,7 +23,7 @@ jobs:                                                      -0.14
                                                                           +0.00
     steps:                                                                -0.03
       - uses: "actions/checkout@v2"                                       +0.02
-      - uses: "actions/setup-python@v1"                                   -0.04
+      - uses: "actions/setup-python@v2"                                   +0.11
         with:                                                             +0.03
           python-version: "${{ matrix.python-version }}"                  -0.28
       - name: "Install dependencies"                                      -0.22
```

Collateral | true: misaligned / collateral | P(misaligned) = 0.957 | attrs, SIFT-017393

Intent (word-level IG)

Bump CodSpeedHQ / action from 3 . 5 . 0 to 3 . 8 . 0 . Bumps [CodSpeedHQ / action] (https : ...

Hunk lines (line-level IG: sum of token attributions per line)

```
src/attr/_funcs.py                                                        -0.20
@@ -377,11 +377,9 @@ def resolve_types(cls, globalns=None, localns=None, attribs=None):  -0.30
     ..  versionadded:: 21.1.0 *attribs*                                   -0.05
                                                                           +0.00
     """                                                                   -0.01
-    try:                                                                  +0.01
-        # Since calling get_type_hints is expensive we cache whether we've +0.02
-        # done it already.                                                +0.03
-        cls.__attrs_types_resolved__                                      +0.04
-    except AttributeError:                                                +0.00
+    # Since calling get_type_hints is expensive we cache whether we've    +0.03
+    # done it already.                                                    +0.04
+    if getattr(cls, "__attrs_types_resolved__", None) != cls:             +0.17
         import typing                                                     +0.02
                                                                           +0.00
         hints = typing.get_type_hints(cls, globalns=globalns, localns=localns) +0.17
```

Scope-Expansion | true: misaligned / scope_expansion | P(misaligned) = 0.898 | axios, SIFT-017837

Intent (word-level IG)

: Send ` false ` flag isStandardBrowserEnv for Nativescript

Hunk lines (line-level IG: sum of token attributions per line)

```
lib/utils.js                                                              +0.18
@@ -18,6 +18,15 @@ function isArray(val) {                                    -0.15
   return toString.call(val) === '[object Array]';                         -0.11
 }                                                                         +0.03
                                                                           +0.03
+/**                                                                       +0.05
+ * Determine if a value is undefined                                      +0.10
+ *                                                                       +0.02
+ * @param {Object} val The value to test                                  -0.52
+ * @returns {boolean} True if the value is undefined, otherwise false    -0.03
+ */                                                                       +0.02
+function isUndefined(val) {                                               +0.17
+  return typeof val === 'undefined';                                      +0.20
+}                                                                         -0.02
                                                                           +0.00
 /**                                                                       +0.74
  * Determine if a value is a Buffer                                       +0.24
```

Purpose-Drift | true: misaligned / purpose_drift | P(misaligned) = 0.908 | httpx, SIFT-017387

Intent (word-level IG)

Bump the python - packages group with 8 updates

Hunk lines (line-level IG: sum of token attributions per line)

```
tests/conftest.py                                                         -0.13
@@ -103,6 +105,16 @@ async def slow_response(scope, receive, send):              +0.19
     await send({"type": "http.response.body", "body": b"Hello, world!"})  +0.07
                                                                           +0.00
                                                                           +0.01
+async def premature_close(scope, receive, send):                         +0.03
+    await send(                                                          +0.02
+        {                                                                +0.02
+            "type": "http.response.start",                               +0.22
+            "status": 200,                                               +0.06
+            "headers": [[b"content-type", b"text/plain"]],               +0.04
+        }                                                                +0.02
+    )                                                                    +0.04
+                                                                         +0.02
+                                                                         +0.02
 async def status_code(scope, receive, send):                             +0.13
     status_code = int(scope["path"].replace("/status/", ""))             +0.16
     await send(                                                          +0.01
```

Figures 2-5: the four cases (summary table above). In the Aligned case the model gives low weight to any single line; the misaligned cases show scattered positive scores over several lines rather than one decisive line, so the highlights should be read as where the model looked, not as a verdict on which line is the unrequested change.

7. Faithfulness Check

The deletion test replaces the top-k attributed tokens by [PAD] and measures the drop in P(misaligned); a faithful attribution should drop it more than masking k random tokens.

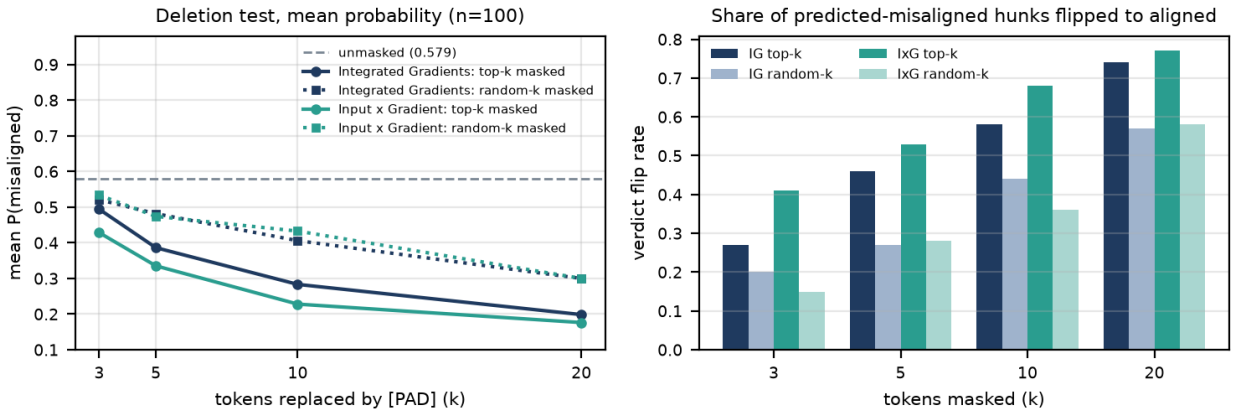


Figure 6. Mean P(misaligned) after masking k tokens, and the share of hunks whose verdict flips to aligned (n = 100 test hunks).

k	IG top-k	IG random-k	IG flip top / random	IxG top-k	IxG random-k	IxG flip top / random	IG top beats random
3	0.494	0.521	0.27 / 0.20	0.428	0.534	0.41 / 0.15	52%
5	0.386	0.482	0.46 / 0.27	0.335	0.474	0.53 / 0.28	66%
10	0.283	0.406	0.58 / 0.44	0.228	0.433	0.68 / 0.36	64%
20	0.198	0.300	0.74 / 0.57	0.176	0.300	0.77 / 0.58	64%

Mean unmasked P(misaligned) is 0.579. Top-k masking always lowers it more than random masking, with the clearest gap at k = 5 to 10 (IG k = 5: drop 0.192 vs 0.097; IxG k = 5: 0.243 vs 0.105). At k = 3 the gap is small (0.084 vs 0.058) and the top set beats random in only 52% of hunks. Input x Gradient separates from random slightly more than IG here, so IG is not shown to be the more faithful method by this test.

8. Cost of Explanation

Measure	Value	Setting
Scoring one hunk (no explanation)	p50 24.7 ms	cuda, NVIDIA GeForce RTX 4050 Laptop GPU
IG explanation, one hunk	p50 506 ms / p95 1328 ms	24 steps, bf16
Ratio explain / score (p50)	20.5x	computed
Peak GPU memory during XAI run	0.99 GB	capped at 1.5 GB
Explanation latency on CPU	-	the UI marks it as slow on CPU

Source note in the JSON: may be inflated if other jobs shared the GPU. IG needs 24 forward and backward passes, which explains the order-of-magnitude cost; the UI uses 16 steps, so its cost is proportionally lower.

9. Scope and Limitations

- Attribution is evidence about what the model used, not proof of what makes a hunk misaligned; the four cases are favourable examples chosen by confidence.
- [PAD] masking is itself off-distribution: random masking alone lowers P(misaligned) from 0.579 to 0.482 at k = 5, so only the gap to random is informative, and a different replacement (mean embedding, token deletion) was not tested.
- Faithfulness uses n = 100 hunks from one split and one trained model (single seed (473), no confidence intervals); no confidence interval is given.
- The deletion test measures sufficiency of the top tokens for the prediction, not whether a human finds the highlight useful; no user study was done.
- Labels in Sift-IA v1.0 are construction-determined (a misaligned hunk is a donor hunk from a different commit); there is no human validation.

- Latency and size come only from the machine they were measured on (Intel i7-13650HX CPU, RTX 4050 Laptop GPU with 6 GB); every run is a single seed (473), no confidence intervals.

10. Summary of Contributions

Implemented IG and Input x Gradient on the cross-encoder with intent-side and hunk-side aggregation, integrated the explanation into the review UI, and measured faithfulness and cost on the test split. The methods (Integrated Gradients, Input x Gradient, deletion tests) are existing techniques; the contribution here is applying them to the hunk-versus-intent task and surfacing them in the review workflow. Everything above is regenerated from results/xai.json and results/xai_cases_ui.json by experiments/build_reports.py.